DICOM Header Auto Repair

User Manual

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1. Overview

The DICOM Header Auto Repair tool is designed to identify and correct common errors in DICOM file headers that prevent successful import into medical imaging systems such as Eclipse. DICOM (Digital Imaging and Communications in Medicine) files contain metadata headers that must conform to strict standards for proper functionality.

This application automates the detection and repair of frequent header issues, significantly reducing manual intervention time and improving workflow efficiency.

2. System Requirements

- Windows 7 or later

- Python 3.7 or later
- At least 500 MB of free disk space
- Read/write access to source and destination directories

3. Getting Started

Installation

1. Extract the application files to your desired directory
2. Run the included virtual environment batch file: `MakeVenv.bat`
3. Wait for the installation to complete

Launching the Application

Double-click the `DICOM Repair.bat` file or the `DICOM Repair` desktop shortcut to launch the application. The GUI will appear within a few seconds.

4. Using the DICOM Repair Tool

Main Interface

The DICOM Repair tool features a straightforward graphical interface with the following components:

DICOM Repair Tool Main Interface

Figure 1: DICOM Repair Tool Main Interface

> **Input Directory Selection**
> Select the directory containing DICOM files to be repaired. Click the **Browse** button to open a file dialog.

> **Include Sub-Directories**
> Check this option to recursively process all DICOM files in subdirectories of the selected

> input directory.

> **Output Directory Selection**
> Select where repaired DICOM files will be saved. Click the **Browse** button to open a file dialog.

Step-by-Step Repair Process

1. **Select Input Directory:**
   - Click the **Browse** button next to "Select the directory containing DICOM files to be repaired"
   - Navigate to and select the folder containing your DICOM files
2. **Configure Processing:**
   - Check "Include Sub-Directories?" if your DICOM files are organized in subfolders
3. **Select Output Directory:**
   - Click the **Browse** button next to "Select the directory where repaired DICOM files will be saved"
   - Choose or create a folder where repaired files should be saved
4. **Start Repair:**
   - Click the **Repair** button to begin processing
   - The status panel will display progress information
5. **Monitor Progress:**
   - The status section displays real-time information about files being processed
   - A cyan progress bar at the bottom of the window indicates the overall completion percentage
   - Once complete, the status will show a summary of repairs made

Repair in Progress

Figure 2: Repair Process in Progress

Completed Repair

Figure 3: Repair Completed

> **Tip:** You can exit the application at any time using the **Quit** button or the window close button. However, it is recommended to allow the repair process to complete.

5. Repair Functions

The DICOM Header Auto Repair tool automatically detects and corrects the following types of issues:

Automatic Repairs Performed

- **Invalid Character Removal:** Eliminates non-printable or invalid characters from string fields
- **Modality Correction:** Verifies and corrects incorrect DICOM modality tags
- **Address Standardization:** Resolves mismatched institution addresses across related series
- **Tag Validation:** Ensures all critical DICOM tags conform to standards

Output Files

Repaired DICOM files are saved to the specified output directory with the same filenames as the originals. Original files remain unmodified.

6. Troubleshooting

Application Won't Start

- Ensure Python 3.7 or later is installed
- Run the `MakeVenv.bat` file to install required dependencies
- Check that you have adequate disk space

No Files Are Being Processed

- Verify that the input directory contains valid DICOM files (usually with .dcm extension)
- Check that "Include Sub-Directories?" is selected if files are in subdirectories

- Ensure you have read permissions on the input directory

Output Directory Access Issues

- Verify that you have write permissions on the output directory
- Ensure the output directory has sufficient free space
- Try using a different output directory if the current one is inaccessible

Repairs Don't Resolve Import Issues

If import issues persist after running the DICOM Repair tool, refer to Appendix A for detailed information about the automatically repaired problems. If you encounter a new type of issue not covered in this manual, see the **DICOM Repair Maintenance & Deployment Guide** for instructions on manually investigating DICOM headers and extending the tool with custom repairs.

Appendix A: Known Problems and Automatic Repairs

This appendix describes common DICOM import problems that are automatically detected and repaired by the DICOM Header Auto Repair tool. For each problem, you will find the symptoms, diagnosis, epidemiology, and the automatic repair applied by the tool.

Problem A: Incorrect Image Modality

Symptoms

- The images can be imported by Varian Eclipse, but Contouring crashes when the image set is opened

Diagnosis

- The DICOM Modality tag `(0008, 0060)` of the PET image series is labeled as "CT"
- In the past there have also been cases where CT image sets have been incorrectly labeled as 'NM' (Nuclear Medicine) or 'OT' (Other)

Epidemiology

- Recently, the most common case has been where PET images have been given the Modality 'CT'
- This occurs infrequently, but in short bursts
- The problem is usually resolved on the sending end within 1 day
- When one is found, watch for other image sets sent from the same institution within a 1-2 day time window

Automatic Repair

1. The DICOM Repair tool automatically corrects the DICOM Modality tag `(0008, 0060)` to the appropriate value:
   - 'PT' for Positron Emission Tomography (PET) images
   - 'CT' for Computed Tomography images
2. Note: If images with the incorrect modality have already been imported into Eclipse, they must be manually deleted from the "Selection" workspace before re-importing the repaired files
3. After running the repair tool, re-import the corrected DICOM files into Eclipse

Problem B: Invalid Characters

Symptoms

- The images cannot be imported
- Import error message states: "Value is invalid: String contains '\' ...."
- Alternate import error message: "The patient of the data to be imported is unidentified or none of the data could be imported"

Diagnosis

- The DICOM "Body Part Examined" tag `(0018, 0015)` in the image series begins with a forward slash "/"
- In the past there have also been cases where the DICOM "Other Patient IDs" tag `(0010, 1000)` (no longer in use) began with a "/"
- Technically, any String DICOM value beginning with a "/" could cause this issue
- **Note:** The "/" is actually not present in the header; this is MicroDicom's notation indicating the value contains non-printable characters

Epidemiology

- Recently, this problem has resulted from "illegal" characters in the CT BodyPart DICOM header tag
- In the past, cases have occurred in the "Other Patient IDs" tag `(0010, 1000)`, but this tag is no longer in use, making it unlikely
- These problems are relatively common
- Almost all cases have been in image sets coming from Ottawa, though not all images from Ottawa have this problem

Automatic Repair

1. The DICOM Repair tool automatically identifies and removes non-printable and invalid characters from all string fields
2. Specifically, any characters that would prevent successful import (such as control characters or invalid Unicode sequences) are cleared
3. After running the repair tool, re-import the corrected DICOM files into Eclipse

Problem C: Mismatched Addresses

Symptoms

- Varian Eclipse crashes while attempting to import the images
- Import error message includes the text: "Cannot insert duplicate key row in object 'dbo.Equipment'"

Diagnosis

The DICOM "InstitutionAddress" tag `(0008, 0081)` is different for the CT and PET images, while other "Device References" remain the same:

DICOM Tag	Tag Number
Manufacturer	(0008, 0070)
Institution Name	(0008, 0080)
Station Name	(0008, 1010)
Institutional Department Name	(0008, 1040)
Manufacturer Model Name	(0008, 1090)
Device Serial Number	(0018, 1000)
Software Versions	(0018, 1020)

Epidemiology

- This error is unique to PET image sets coming from Mississauga and occurs with all PET image sets from that location
- This is so common that when diagnosing an import issue, the first step is to check whether it came from Mississauga, and if so, proceed assuming this is the problem
- **Technical Background:** This problem arises from a known issue in the way Varian uniquely identifies diagnostic equipment in its database:
  - The DICOM Import/Export application checks the uniqueness of all equipment values

- However, the dbo.Equipment table's index XAK1Equipment checks all equipment values except the InstitutionAddress
- Therefore, if only the InstitutionAddress is different during the insertion of a new entry in the dbo.Equipment table, the operation fails and throws an error as it is considered a duplicate entry

Automatic Repair

1. The DICOM Repair tool automatically detects when the InstitutionAddress `(0008, 0081)` differs between related image series (CT and PET)
2. The tool clears the InstitutionAddress value while preserving all other device reference information
3. This ensures that all related images are identified as part of the same equipment setup, preventing the duplicate key error in Eclipse
4. After running the repair tool, re-import the corrected DICOM files into Eclipse

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For additional information, please refer to the application README.md file or contact your IT support team.